



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM



Introduction to FastAPI



PRIT

CS-CLUB





INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM



Before FastAPI, What is an API?





API: Application Programming Interface

API (Application Programming Interface) is a **set of rules** that allows **one software program to interact with another.**

It defines:

- **How to request data or perform actions**
- **What kind of data to send**
- **What kind of response to expect**





What is REST and RESTAPI?

REST stands for **Representational State Transfer**.

It's a way of designing how two systems (like a website and a server) should talk to each other using the internet.

- Think of it like a set of rules that websites and apps follow when they want to exchange data.
- A **REST API** is an API that follows the **REST** rules. It lets one app talk to another using standard **HTTP methods**.





What is RESTFUL Service?

A RESTful service is just a web service that follows the REST style.

It helps apps talk to each other over the internet by sending and receiving data using URLs and HTTP methods.



API vs RESTAPI

<u>Aspect</u>	<u>API</u>	<u>REST API</u>
<u>Meaning</u>	A general way for two apps to talk to each other	A type of API that follows REST rules
<u>Usage</u>	Can work over many protocols (HTTP, FTP, etc.)	Works only over HTTP
<u>Rules</u>	No specific structure	Follows REST architecture (uses HTTP methods like GET, POST)
<u>Data Format</u>	Can use any format (XML, JSON, etc.)	Mostly uses JSON or XML





HTTP Methods

HTTP methods are like commands used by your app to talk to a server over the internet.

HTTP methods: GET, POST, PUT and DELETE

HTTP Methods

GET: for reading the data

POST: to create or append new data

PUT: to update existing data

DELETE: to delete existing data

GET	/pet/{petId}	Find pet by ID
PUT	/pet	Update an existing pet
DELETE	/pet/{petId}	Deletes a pet
POST	/pet/{petId}/uploadImage	uploads an image



What is FastAPI and Uvicorn

FastAPI is a modern, fast (high-performance) web framework for building APIs with Python 3.6+.

- It helps you create REST APIs easily — with less code, auto validation, and even automatic docs using Swagger UI.

Uvicorn: Uvicorn is a lightweight web server used to run FastAPI apps.





Benefits of FastAPI

- 1.) Fast and Efficient:** Handles many requests quickly — great performance for modern web apps.
- 2.) Less Code, Easy to Read:** Clean syntax using Python type hints — beginners find it friendly.
- 3.) Automatic Data Validation:** If you expect an integer and get a string, FastAPI will auto-handle the error.
- 4.) Built-in API Docs (Swagger UI):** FastAPI creates interactive docs automatically — no extra work.



Installation

For Mac/Linux users:

```
# Create a virtual environment (optional but recommended)
python3 -m venv venv
source venv/bin/activate

# Install FastAPI and Uvicorn
pip install fastapi uvicorn
```



Installation

For Windows :

```
:: Create a virtual environment  
python -m venv venv  
  
:: Activate the virtual environment  
venv\Scripts\activate  
  
:: Install FastAPI and Uvicorn  
pip install fastapi uvicorn
```





Run command and Live Demo

Run command:

```
uvicorn main:app --reload
```




Run command and Live Demo

Run command:

```
uvicorn main:app --reload
```



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM



Any Questions?





Take Home Assignment

Task: Make a to-do list app

Goals: implement all http methods

Test it out in SwaggerUI

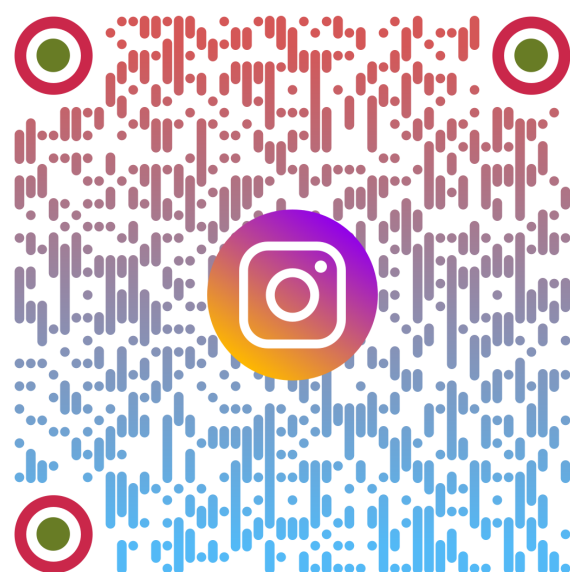
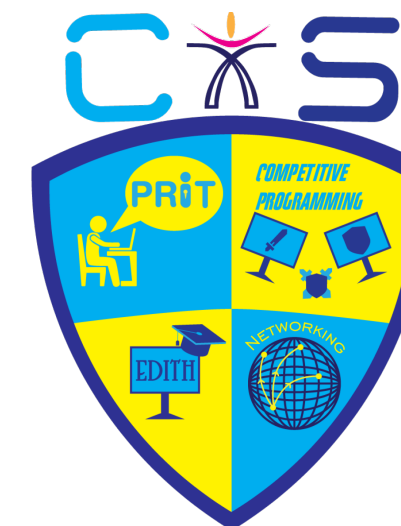
Hint: Refer to the students demo code



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM

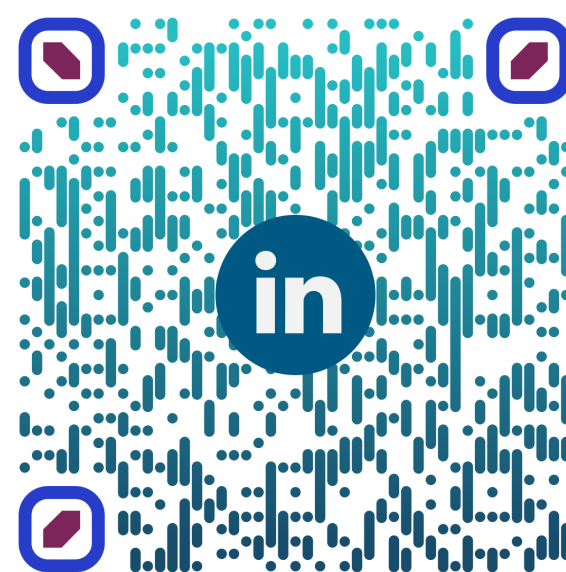


Connect with us



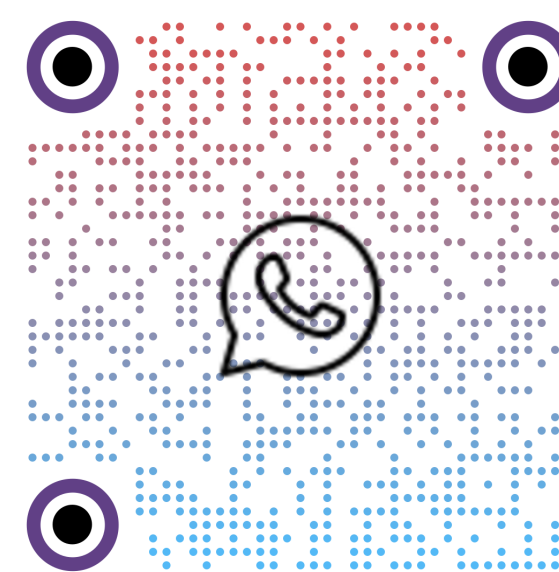
cs.club.

iiitdm

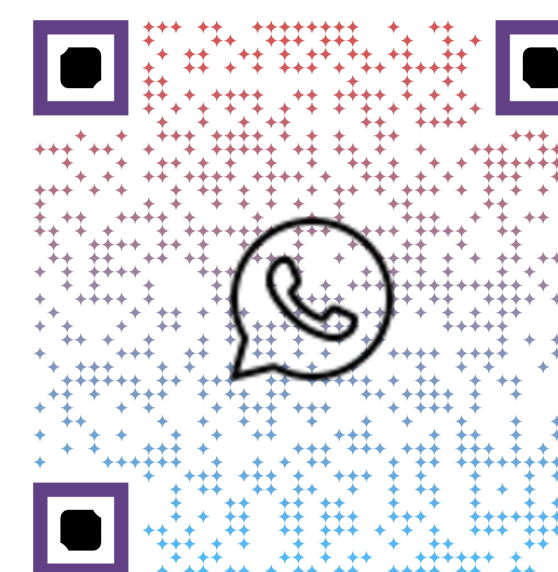


CS Club

IIITDM



Tech Affairs WhatsApp Channel



Ignite WhatsApp Group



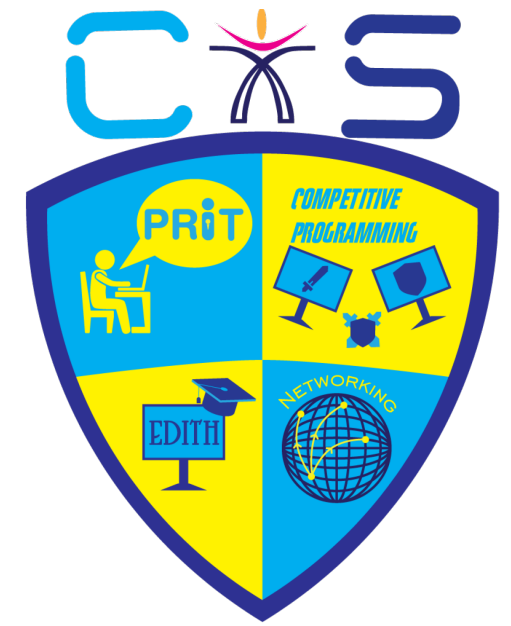
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM



Scan for Video



CS Club IIITDM

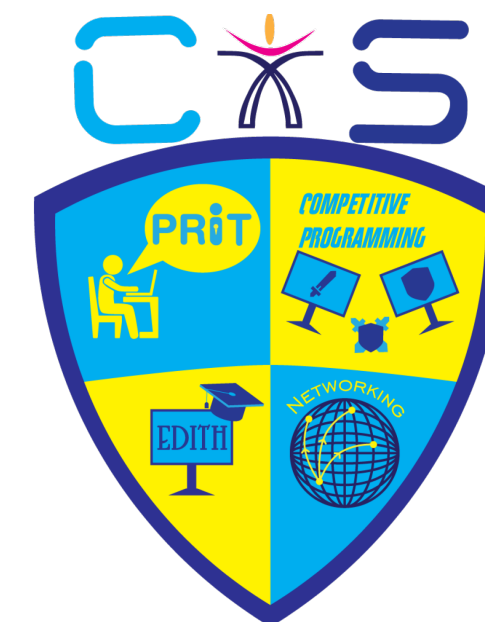




INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM



Thank You



Share your Feedback about this session





INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM



Surprise Revelation

Become an Ignite Brand Ambassador!

Share a short video (2-3 min in length)  about your experience with Ignite Sessions.

Top entries will be featured on our Instagram & YouTube 

Chance to become our official Brand Ambassadors 

Scan the QR below to fill your entry.

